

Dflare AI

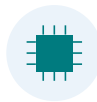
Data Sheet

Modern AI workloads require massive GPU scale, high-throughput data pipelines, strict multi-tenant isolation, and support for both cloud-native and HPC workloads. Traditional cloud and on-prem systems fail to deliver all four simultaneously. Traditional cloud providers impose virtualization overhead and noisy-neighbor effects that degrade training throughput, while on-premises HPC clusters lack the self-service provisioning and lifecycle management that modern AI teams require. Dflare AI was purpose-built to address this gap — providing unified GPU infrastructure that combines bare metal performance, hardware-enforced isolation, and full lifecycle automation.

Dflare AI

An enterprise GPU infrastructure platform designed to deliver bare metal performance with cloud-like usability. The platform is composed of three primary layers: Access Layer (Portal UI, REST APIs, CLI), Control Plane (Workflow Orchestrator, Cluster Manager, Network Manager, Identity & Access, Monitoring & Metering), and Data Plane (GPU nodes, Kubernetes / Slurm workloads, InfiniBand fabric, Parallel filesystem). Dflare AI efficiently manages bare metal GPU nodes across any infrastructure, offering a unified experience for application and infrastructure lifecycle management. The platform caters to enterprises, AI labs, managed service providers, and government organizations, assisting in GPU provisioning, workload orchestration, security, and billing.

Industry Use Cases



Large-Scale AI Training: Multi-node, multi-GPU distributed training across hundreds of GPUs with RDMA-based InfiniBand interconnect. Scale AI training from 8 to 10,000+ GPUs without architectural changes, achieving near-linear scaling.



Unified Kubernetes + Slurm Orchestration: Run both containerized and Slurm workloads on the same bare metal infrastructure with unified networking, storage, security, and billing — managed through a single control plane.



GPU-as-a-Service for Multi-Tenant Environments: Offer dedicated bare metal GPU clusters to multiple tenants with hardware-level isolation at InfiniBand switch (partition key), filesystem (access control map), and network fabric (VRF/VXLAN).

Key Features



Bare Metal GPU Performance:

Direct GPU access without virtualization overhead. Hardware-level BIOS and OS tuning pre-applied via golden images. Near-native efficiency (99%+ of bare metal GPU peak).



Dual-Fabric Network Architecture:

Ethernet (VXLAN/EVPN/BGP) for control plane and tenant VPCs. InfiniBand for RDMA-based high-performance GPU-to-GPU and GPU-to-storage communication with Partition Keys for tenant isolation.



Central IAM and Multi-Tenancy Controls:

Integrates with enterprise identity providers (OIDC/OAuth2). Issues short-lived JWTs, enforces RBAC/ABAC at every API boundary, and isolates tenants via dedicated realms and scoped tokens for enhanced **multi-tenancy support**.



Observability:

GPU utilization, cluster health, and job performance monitoring. Metrics collectors, time-series database, and dashboards for real-time observability across compute, network, and storage layers.



Zero Trust Security:

Defense-in-depth security model. OAuth2/OIDC identity, RBAC + ABAC authorization, VRF + VLAN + PKey network isolation, storage ACLs, and TLS 1.2+ / mTLS encrypted service-to-service transport.



Unified Kubernetes + Slurm:

Single control plane manages Kubernetes and Slurm. Kubernetes handles containerized workloads via device plugins. Slurm handles batch workloads with GPU-aware scheduling using GRES. Both share the same underlying nodes, storage, and network.

Key Benefits



Automated Lifecycle Management:

From bare metal power-on to production cluster — fully automated. No SSH, no manual configuration. Provision → operate → monitor → bill.



Hardware-Level Tenant Isolation:

Isolation at InfiniBand switch hardware (partition key), filesystem (access control map), and network fabric (VRF/VXLAN). Enforce strict multi-tenancy through hardware and software controls.



Scalable Multi-Cluster Management:

Scale AI training from 8 to 10,000+ GPUs without architectural changes. Horizontally scalable GPU nodes with leaf-spine fabric expansion.



Near-Bare-Metal Performance:

Eliminate virtualization overhead — achieve near-bare-metal performance with cloud-like operational simplicity. Zero hypervisor, direct device access.



Comprehensive Observability:

Real-time observability across compute, network, and storage layers enables proactive capacity management and rapid incident response.



Zero Trust Security:

Defense-in-depth security model based on zero-trust principles ensures a high level of security for the managed GPU infrastructure.

For more information or questions about Coredge's Dflare AI:



Please visit <https://coreedge.io>



Contact us at info@coreedge.io